

Text-Guided Image Clustering with Unsupervised Dual-Level Knowledge Distillation

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Abstract—The performance of deep clustering is often limited by the scale and representational capacity of self-supervised models. Recent research has explored the use of large-scale models to mine external semantic information for guiding image clustering. However, these methods often overlook the deployment challenges posed by the substantial number of parameters in such models, which hinders their practical application in resource-constrained environments. In this paper, we propose a novel text-guided unsupervised dual-level knowledge distillation clustering framework, which transfers knowledge from large models to lightweight models. This framework addresses the performance bottleneck of deep clustering and reduces the computational and storage costs associated with the deployment of large-scale models. Our method consists of two key phases: i) In the clustering phase, the vision language model is used to construct an external text space, and high-quality image clustering is achieved by combining cross-modal consistency learning of images and texts. ii) In the distillation phase, feature distillation and cluster distillation transfer knowledge from the teacher model in the clustering phase to a smaller student model, significantly reducing the model size while maintaining clustering performance. Extensive experiments on four common benchmark datasets and two complex scene datasets demonstrate the effectiveness of the proposed method in both clustering and distillation. In particular, our method achieves the best clustering performance, using only 1/44 of the parameters of the teacher model.

Index Terms—Machine learning, deep learning, image clustering, knowledge distillation, vision language model.

I. INTRODUCTION

As a core task of data mining and machine learning, cluster analysis has long been a focus of extensive research. With the rapid advancement of information technology, the data we encounter today has become unprecedentedly large-scale, diverse, and high-dimensional. These data include images, texts, social media information, and multi-view, multi-modal data from various sensor devices, all of which are growing in scale and complexity. In this context, clustering technology has become an important tool for processing and understanding such data. The primary objective of clustering is to partition a

set of unlabeled data into several groups based on similarity or correlation, ensuring that data within the same group exhibit high similarity, while data between different groups have low similarity, thereby revealing the inherent structure and potential patterns of the data. Clustering is widely applied across various fields, including computer vision [1]–[7], natural language processing [8], [9], social network analysis [10]–[12], and biomedicine [13]–[15].

Deep clustering [16]–[19], leveraging the powerful representation learning capabilities of deep neural networks, has demonstrated excellent performance in processing high-dimensional, complex datasets such as images. In recent years, researchers have designed various self-supervised learning tasks, including feature reconstruction [20], [21], contrastive learning [19], [22], [23], neighborhood consistency learning [24], [25], and pseudo-labeling learning [26]–[28], to improve clustering performance. These methods rely on latent information within the data to learn more discriminative representations unsupervised, significantly advancing the field of deep clustering. However, mining self-supervised information often faces challenges, particularly those related to model capacity and training strategies. To address these issues, recent methods mine external prior information to guide image clustering, demonstrating strong potential and garnering significant attention.

Externally guided deep clustering methods typically leverage vision language models (VLMs) to construct an auxiliary text space, enabling image clustering through text guidance. Based on how text space is constructed, these methods can be categorized into two paradigms: matching-based [29]–[33] and generative-based methods [34], [35]. As shown in the Fig. 1. Matching-based methods align visual representations with predefined or retrieved textual semantics, while generative-based construct the text space by generating image captions, thus converting image clustering into efficient text clustering. Regardless of the paradigm, the quality of the constructed textual space directly affects image clustering performance. These methods leverage pre-trained VLMs as semantic bridges between image and text representations, providing reliable semantic information for clustering. Although these methods have made notable advancements in clustering performance, their reliance on large-scale VLMs introduces substantial computational and deployment challenges, especially in resource-constrained environments. More importantly, such models implicitly contain rich semantic knowledge that bridges visual and textual modalities, yet how to transfer this knowledge to a lightweight model in an unsupervised clustering scenario remains underexplored. Unlike existing methods that focus on improving clustering performance, this work addresses

This work was supported by the National Natural Science Foundation of China (62376252, 92470202); Key Project of Natural Science Foundation of Zhejiang Province (LZ22F030003); Zhejiang Province Leading Geese Plan (2025C02025, 2025C01056); Zhejiang Province Province-Land Synergy Program (2025SDXT004-3); Frontier Technologies R&D Program of Jiangsu (BF2024070).

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clustering-oriented model compression via cross-model distillation, which introduces new challenges in transferring semantic and structural knowledge without supervision. Therefore, an essential motivation of this work is to enable efficient clustering by distilling multi-modal semantic knowledge from a large teacher model into a compact student model while significantly reducing inference cost.

In this study, we proposed a text-guided Unsupervised dual-level Knowledge Distillation Clustering (UKDC) method, which aims to address the performance limitations of lightweight models in self-supervised clustering tasks with the computational and resource demands of large models in practical deployment. Specifically, our method consists of two phases: the clustering phase and the distillation phase. In the clustering phase, we designed different text space construction strategies to generate image captions consistent with the clustering objectives. Cross-modal consistency learning is then employed to jointly embedded features of image and text, resulting in high-quality image cluster assignments. In the distillation phase, the model from the clustering phase is used as the teacher model, and the cluster assignments are used as target distribution. Through a dual-level knowledge distillation, the features and cluster distributions between the student and the teacher are effectively aligned. Text guidance plays a dual role in our framework, encompassing both text space construction and cross-modal knowledge distillation. In the clustering phase, we interact with a VLM through customized prompts, guiding the model to generate image captions that align with the clustering objectives in VQA manner. In the knowledge distillation phase, the textual representations produced by the teacher model serve as semantic guidance signals, enabling the student model to acquire high-level semantic knowledge while alleviating redundancy in visual representations. The major contributions of this work are summarized as follows:

- We propose UKDC, a novel unsupervised clustering framework that introduces an explicit cross-model teacher-student paradigm into deep clustering, enabling multi-modal semantic knowledge transfer from a large-scale VLM to a lightweight student model without any ground-truth labels.
- We design two complementary text space construction strategies — a general strategy for standard benchmarks and a customized strategy for complex scenes — to address the semantic misalignment and intra-text inconsistency inherent in existing matching-based methods.
- We propose a dual-level unsupervised distillation mechanism for clustering-oriented model compression, which jointly performs feature-level and cluster-level knowledge transfer, enabling up to 44× model compression while preserving over 88% clustering performance.
- On four common benchmark datasets and two complex scene datasets, the clustering performance of our method surpasses the SOTA before and after distillation, fully verifying the effectiveness and generalization ability of the proposed deep clustering and distillation mechanisms.

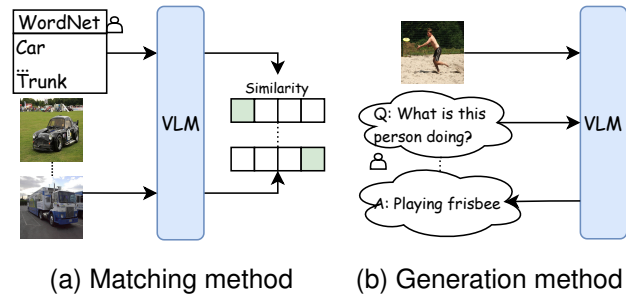


Fig. 1: **Comparison of text construction methods.** The matching method selects nouns by the similarity between the image feature and the noun feature. The generation method directly generates the caption corresponding to the image.

II. RELATED WORKS

A. Deep Clustering

According to the source of information, deep clustering methods can be divided into two categories: internal information mining and external information guidance. Internal information mining methods rely on the data inherent structural information, typically by designing pretext tasks to improve feature representation. For instance, DEC [20] learns the latent embedding space by reconstruction of auto-encoder. SCAN [24] achieves clustering by learning transformation invariance and combining neighborhood consistency. NNM [25] further optimizes features by combining global and local neighbors. CC [36] introduces both instance-level and cluster-level contrastive learning to enhance feature and cluster discriminability. SPICE [26] constructs prototypes from high-confidence samples and guides neighbor labeling. LFSS [37] uses the stability of samples during training as a supervision signal to improve the performance of deep clustering. Although these methods effectively exploit the internal structure of the data, the lack of semantic guidance makes it difficult to further enhance the quality of feature representations.

Recent studies [29]–[31], [33], [38] have explored the use of external semantic knowledge to guide image clustering. Among matching-based methods, SIC [29] constructs a text space by matching visual cluster centers with predefined nouns and combines consistency learning to improve clustering performance. TAC [31] further enhances semantic representation by filtering and retrieving the most discriminative nouns in images, combined with neighborhood consistency learning for high-quality clustering. MCA [30] optimizes the semantic space construction and introduces a multi-level cross-modal alignment mechanism to strengthen the collaborative representation of image and text features. However, matching-based methods rely on the static matching and filtering of predefined nouns [39], which inevitably leads to intra-text inconsistency and text-image semantic misalignment. Generation-based methods [34], [35], [40], on the other hand, use text generation models to construct semantic spaces. While these methods avoid the above issues, the reliance on a single text modality still leads to suboptimal clustering results.

In contrast, our proposed method builds upon and enhances these approaches. We introduce two caption generation strategies to construct a more flexible text space, and adopting cross-modal consistency learning to align text and image cluster spaces, fully utilizing both text and image information.

B. Vision Language Pre-training

In recent years, self-supervised learning methods [22], [23], [41] have led to significant breakthroughs in the feature representation of high-dimensional visual data. Building upon this, VLMs that incorporate language information have further enhanced the semantic capabilities of image representations, advancing the field of multi-modal learning. Notable works, such as CLIP [42] and ALIGN [43], train image encoders on large-scale image-text pairs within a contrastive learning framework, achieving remarkable improvements in tasks like image retrieval and zero-shot classification. Generative models, including BLIP [44] and BLIP-2 [45], combine image matching and generation capabilities, further enhancing multimodal understanding. The focus then shifted toward Visual Instruction Tuning with the emergence of LLaVA [46], which demonstrated that simple projection layers combined with high-quality instruction data could transform large-language models into capable multimodal assistants. Currently, the landscape is dominated by native multimodal models like the Qwen3-VL [47] and InternVL [48] series, which integrate "thinking modes" for complex reasoning, Mixture-of-Experts for computational efficiency, and high-resolution dynamic encoding. Modern research is now pushing toward agentic capabilities, where models not only perceive and reason but also act within digital and physical environments through long-context video understanding and fine-grained spatial grounding.

For a fair comparison, this paper uses BLIP-2 as the text generator and CLIP as the text and image encoder in the clustering phase.

C. Knowledge Distillation

Knowledge distillation is a commonly used model compression [49] technique that aims to effectively transfer the knowledge contained in the large and complex teacher model to the lightweight student model. Hinton et al. [50] first propose logit distillation method to imitate the teacher's soft label distribution by minimizing the KL-divergence between the classification outputs of the two models. FitNets [51] introduces the concept of "hints"—transferring intermediate representations from the teacher to guide the student's learning process. Subsequent works have extended KD to self-supervised and contrastive learning paradigms. CRD [52] leverages contrastive objectives to maximize the agreement between teacher and student representations. Similarly, SSKD [53] combine knowledge distillation with self-supervised learning, using pretext tasks to enhance representation transfer. SEED [54] proposes a method to directly distill momentum teacher features, which achieves large model compression without labels and efficiently obtains powerful small models. CLIP-CID [55] leverages cluster-instance discrimination to facilitate knowledge transfer from the teacher model to the student

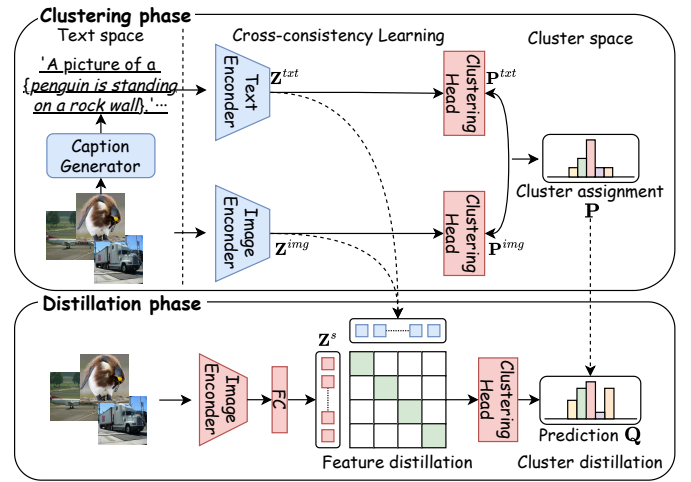


Fig. 2: **UKDC framework pipeline.** Clustering phase includes a caption generator, a text-image encoder, and two clustering heads. The text generator generates captions corresponding to the images, and the text and image encoder extracts the embedding features of the text and image, respectively. Through cross-modal consistency learning, the image is mapped to the final cluster assignment. Distillation phase includes an encoder, a fully connected layer, and a clustering head. The fully connected layer is used to align the feature dimensions between the student model and the teacher model, and combines the features and cluster information of the previous phase to achieve dual-level unsupervised distillation and promote the learning of the student model. Blue and red represent the freezing and training of the module, respectively.

model, thereby empowering the student model to acquire a holistic semantic comprehension of the pre-training data.

Compared to the above supervised or uni-modal distillation approaches, our proposed UKDC framework operates in an unsupervised clustering setting, which does not rely on any ground-truth labels. It's worth noting that existing deep clustering methods [29]–[31] may incorporate mechanisms resembling distillation (e.g., consistency regularization or self-distillation), but they are primarily designed to improve clustering performance within a single model. In these methods, the backbone is frozen and only the cluster heads are optimized to enhance feature partitioning, without explicitly considering knowledge transfer or model compression between models. In contrast, our approach formulates knowledge distillation as a core component for clustering-oriented model compression, where an explicit cross-model teacher–student paradigm is introduced. A large-scale VLM serves as the teacher to transfer both semantic and structural knowledge to a lightweight student model, enabling efficient deployment while preserving clustering performance.

III. METHOD

A. Preliminaries

Given a set of unlabeled image datasets $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N \in \mathbb{R}^{N \times D}$ and number of categories K , our research objectives

include clustering and distillation. First, a caption generator $\mathcal{G}$ is used to generate corresponding caption for the image, thereby constructing the text space $\mathbf{T}$. Subsequently, the image is divided into K clusters by combining the information of the image space and the text space. Second, after obtaining a high-performance teacher model, we further transfer knowledge to the student model through distillation while retaining the original clustering performance as much as possible. The overall framework is shown in Fig. 2.

B. Text Space Construction Strategies

Externally guided clustering methods rely on high-quality text space construction. Most existing methods use a pair of pre-trained encoders to construct the text space by matching the similarities between text and image embeddings. However, matching-based strategies face two major limitations: intra-text inconsistency and text-image semantic misalignment. First, the overly fine-grained noun matching leads to the generation of semantically similar but textually different nouns. Second, matching-based methods often struggle to effectively focus on clustering objectives when handling complex images, resulting in semantic misalignment between images and texts. To address these limitations, we propose two text construction strategies designed to generate captions with greater semantic quality and adaptability across diverse scenarios.

General Text Space Construction: General strategy directly generates a caption for each image using a pre-trained model, without relying on additional filtering. The quality of the text space primarily depends on the image understanding and language generation capabilities of the model. This strategy is used on common benchmark datasets.

Customized Text Space Construction: For complex scene datasets involving multiple objects, interactions or actions, we adopt a customized strategy to construct a text space aligned with the clustering objectives. In this strategy, users can interact with the pre-trained model in a VQA manner by designing and refining different prompt templates to emphasize specific aspects of the scene, such as human activities or additional target objects, thereby generating more task-relevant textual descriptions. Users can review the generated captions and perform several rounds of interaction until the results meet their requirements. This process requires only minimal human intervention. The customized text space provides more precise semantic guidance and significantly improves clustering performance in complex visual scenarios.

C. Cross-modal Consistency Learning

After constructing the text space, our goal is to achieve efficient image clustering based on the consistency between text and image features. To this end, we first use pre-trained text and image encoders to extract embedding features for both text and images. While directly applying k -means clustering after concatenating the outputs of both encoders yields good results, the feature distributions of different modalities often differ significantly, making it difficult to achieve optimal clustering directly. To address this problem, we introduce a

clustering head at the back end of each encoder to map multi-modal features into cluster space, combined with a consistency learning strategy to ensure the optimal cluster assignment:

$$\mathcal{L}_c = -\log \sum_i^N \mathbf{p}_i^{img} (\mathbf{p}_i^{txt})^\top, \quad (1)$$

where $\mathbf{p}_i$ represents the assignment of the i -th sample.

To prevent clustering collapse, where all samples are assigned to the same cluster, we add a regularization term that constrains the cluster space, ensuring that the samples are uniformly distributed within the cluster space. The regularization term is defined as follows:

$$\mathcal{L}_r = \sum_{j=1}^K \bar{\mathbf{p}}_j^{img} \log \bar{\mathbf{p}}_j^{img} + \bar{\mathbf{p}}_j^{txt} \log \bar{\mathbf{p}}_j^{txt}, \quad (2)$$

$$\text{where } \bar{\mathbf{p}}_j^{img} = \frac{1}{N} \sum_{i=1}^N p_{ij}^{img}, \quad \bar{\mathbf{p}}_j^{txt} = \frac{1}{N} \sum_{i=1}^N p_{ij}^{txt}.$$

The final cluster assignment is represented by the average of the text and image assignments $\mathbf{P} = (\mathbf{P}^{txt} + \mathbf{P}^{img})/2$, $\mathbf{P} \in \mathbb{R}^{N \times K}$.

D. Unsupervised Dual-level Knowledge Distillation

While image clustering methods leveraging VLMs have achieved superior results compared to traditional self-supervised approaches, directly deploying such large-scale models remains challenging in real-world scenarios such as edge computing and industrial applications due to high computational and storage costs. To address these challenges, we propose a dual-level unsupervised distillation clustering framework that transfers multi-modal knowledge from a large teacher model to a lightweight student model, enabling efficient model compression while maintaining performance. In this framework, the feature-level distillation inherits both the teacher's visual information and semantic knowledge to guide the student's representation learning, while the cluster-level distillation inherits the teacher's discriminative ability between clusters, encouraging the student model to preserve high-level cluster separability.

Feature distillation aims to transfer the representation capability of the teacher model to the student model. We employ the InfoNCE [56] loss function for training, treating features from the same sample but encoded by the teacher and student networks as positive pairs, while features from different samples are considered negative pairs. This contrastive learning objective enhances the student model's ability to distinguish instances by maximizing the similarity between positive pairs and minimizing the similarity between negative pairs:

$$\mathcal{L}_{feature}^{img \rightarrow s} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\mathbf{z}_i^s \cdot \mathbf{z}_i^{img} / \tau_f)}{\sum_{j=1, j \neq i}^N \exp(\mathbf{z}_i^s \cdot \mathbf{z}_j^{img} / \tau_f)}, \quad (3)$$

$$\mathcal{L}_{feature} = (\mathcal{L}_{feature}^{img \rightarrow s} + \mathcal{L}_{feature}^{txt \rightarrow s})/2. \quad (4)$$

Here, N denotes the total number of training samples, and i, j indexes each sample pair. $\mathbf{z}_i^s$ and $\mathbf{z}_i^{img}$ represent the

feature embeddings of the i -th sample obtained from the student and teacher image encoder, respectively. Similarly, $\mathbf{z}_i^{txt}$ denotes the text embedding of the i -th sample obtained from teacher text encoder. τ_f is the contrastive temperature control parameter, used to adjust the smoothness of the feature similarity distribution. The term $\mathcal{L}_{feature}^{img \rightarrow s}, \mathcal{L}_{feature}^{txt \rightarrow s}$ represents the distillation loss between the student and teacher networks in intra-modal and inter-modal, respectively. Intra-modal distillation encourages the student model to preserve representation consistency with the teacher in the same modality, facilitating stable knowledge transfer but potentially retaining modality-specific redundant information. In contrast, cross-modal distillation introduces more refined and semantically focused textual information, guiding student models to learn task-relevant discriminative information and reducing category-irrelevant visual redundancy. By jointly leveraging intra-modal and cross-modal distillation, the student model learns representations that are both consistent with teacher model and more discriminative for clustering.

Cluster distillation learns decision boundaries and uncertainty information by matching the logit distributions output by the teacher and student models and cross-entropy loss. In unsupervised clustering, where true labels are unavailable, we use the cluster assignments generated in the clustering phase as the target distribution $\mathbf{P}$ of the teacher model. The student model is then guided to learn the clustering distribution of the teacher model by minimizing the KL-divergence between the prediction distribution $\mathbf{Q}$ and target distribution $\mathbf{P}$:

$$\mathcal{L}_{cluster} = \frac{\tau_c^2}{N} \sum_{i=1}^N \text{KL}(\text{Softmax}(\mathbf{p}_i/\tau_c) \parallel \text{Softmax}(\mathbf{q}_i/\tau_c)), \quad (5)$$

Here, τ_c is the distillation temperature control parameter, used to adjust the smoothness of the cluster space distribution.

E. Training Strategy

Our method consists of two main phases: the clustering phase and the distillation phase.

Clustering phase: In this phase, we construct the text space using VLMs and perform clustering by ensuring cross-modal consistency between text and images. Additionally, feature embeddings and cluster assignments are used as targets for the next training phase. The loss function for the clustering phase is given by:

$$\mathcal{L}_{clustering} = \mathcal{L}_c + \mathcal{L}_r. \quad (6)$$

Distillation phase: In this phase, we use the teacher model features and the cluster assignments obtained in the previous phase as the student model dual-level unsupervised distillation target training. The loss function of the distillation phase is as follows:

$$\mathcal{L}_{distillation} = \mathcal{L}_{feature} + \mathcal{L}_{cluster}. \quad (7)$$

The overall process is summarized in the algorithm 1.

Algorithm 1 UKDC

Input: Dataset $\mathbf{X}$; Number of clusters K ; Temperature parameters τ_f, τ_c ; Clustering epoch T_c and distillation epoch T_d ; Text generator $\mathcal{G}$, Image encoder $\mathcal{E}^I$ and Text encoder $\mathcal{E}^T$; Student model $\mathcal{F}$

Output: Student network and clustering results

Clustering phase:

- 1: $\mathcal{G}.\text{eval}(), \mathcal{E}^I.\text{eval}(), \mathcal{E}^T.\text{eval}()$
- 2: Text space construction $\mathbf{T} = \mathcal{G}(\mathbf{X}, \text{prompt})$
- 3: **for** epoch = 0 to T_c **do**
- 4: Get text features and image features $\mathbf{Z}^{txt} = \mathcal{E}^T(\mathbf{T}), \mathbf{Z}^{img} = \mathcal{E}^I(\mathbf{X})$
- 5: Train the clustering network using Eq. (6)
- 6: **end for**
- 7: Get clustering results and cluster assignments $\mathbf{P}$

Distillation phase:

- 8: $\mathcal{F}.\text{train}()$
- 9: **for** epoch = 0 to T_d **do**
- 10: Get features $\mathbf{Z}^{img}, \mathbf{Z}^{txt}$ and target distribution $\mathbf{P}$ from teacher model
- 11: Get features $\mathbf{Z}^s$ and prediction distribution $\mathbf{Q}$ from student model
- 12: Train the student network using Eq. (7)
- 13: **end for**
- 14: Output student model and clustering results

F. Computational Analysis

To provide a fair evaluation of efficiency, we analyze both the spatial and temporal complexity of our framework. In terms of spatial complexity, the caption generator contains approximately 3,942M parameters, the teacher encoder has 151M parameters, and each clustering head consists of about 1M parameters. In the distillation phase, the student model contains only 21M parameters, achieving substantial model compression. Regarding temporal complexity, the clustering phase mainly performs forward inference for feature and caption extraction and trains only the lightweight clustering heads, making the overall training time negligible. In the distillation stage, Eq. (3) and Eq. (5) run on the student backbone, so the time complexity is the same as that of traditional deep clustering methods, $O(N^2 d_t)$ and $O(NK)$ respectively. Moreover, as shown in Fig. 5, our model converges efficiently, requiring only 100 epochs in total, and it reaches the best performance of conventional deep clustering methods within just 20 epochs.

Although the overall training time of our framework is relatively low, it requires considerable storage resources due to the use of large pre-trained models during the clustering phase. However, this phase can be regarded as an offline auxiliary step performed only once to construct the semantic structure of the teacher model. In practical scenarios, only the distilled student model needs to be deployed, which operates independently of the large-scale teacher components. This allows our approach to deliver superior clustering performance under resource-constrained environments while maintaining a compact model size and low computational cost at inference time.

G. Theoretical Analysis

Feature distillation aims to preserve the mutual information between the teacher and student representations. Let $I(\mathbf{Z}^t; \mathbf{Z}^s)$ denote the mutual information between teacher features $\mathbf{Z}^t$ and student features $\mathbf{Z}^s$. The InfoNCE loss used in Eq. 3 is a lower bound on the mutual information [56], satisfying:

$$I(\mathbf{Z}^t; \mathbf{Z}^s) \geq \log N - \mathcal{L}_{feature}^{img \rightarrow s}. \quad (8)$$

This indicates that minimizing the feature distillation loss directly maximizes the mutual information between teacher and student representations, thereby preserving the discriminative power of the teacher in the student model.

At the cluster distillation, the objective function of KL-divergence constrains the discrepancy between the teacher's and student's predictive distributions:

$$\text{KL}(\mathbf{P} \parallel \mathbf{Q}) = H(\mathbf{P}, \mathbf{Q}) - H(\mathbf{P}), \quad (9)$$

where $H(\mathbf{P})$ and $H(\mathbf{P}, \mathbf{Q})$ denote the entropy and cross-entropy terms, respectively. Since $H(\mathbf{P})$ is constant for a fixed teacher distribution, minimizing Eq. (5) is equivalent to minimizing the cross-entropy $H(\mathbf{P}, \mathbf{Q})$. This optimization forces the student distribution $\mathbf{Q}$ to concentrate probability mass on the same clusters emphasized by the teacher, thereby reducing uncertainty in the student predictions. As a result, when the teacher's cluster assignments $\mathbf{P}$ are sharp, the student is encouraged to produce low-entropy and more confident predictions, leading to entropy-constrained cluster separation. Therefore, the dual-level design jointly preserves the teacher's semantic expressiveness and cluster structure, enabling the student to achieve both semantic consistency and compact cluster separability.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

The datasets used in our evaluation include four common benchmark datasets CIFAR-10, CIFAR-20, STL-10, ImageNet-10 and two complex scene datasets Stanford-40 and PPMI-7. The dataset information is summarized in the Table I.

CIFAR-10 [57] consists of 60,000 32x32 color images in 10 classes, with 6,000 images per class. There are 50,000 training images and 10,000 test images.

CIFAR-20 [57] has 100 classes containing 600 images each. There are 500 training images and 100 testing images per class. The 100 classes in the CIFAR-100 are grouped into 20 superclasses.

STL-10 [58] contains 13,000 labelled images divided into 10 categories, of which 5,000 are used for training and 8,000 for testing. In addition, 100,000 unlabelled images are provided for pre-training.

ImageNet-10 [59] is a subset extracted from the ImageNet-1k [60] dataset containing 10 categories with 1,300 images each.

Stanford-40 Action Dataset [61] contains images of humans performing 40 actions. In each image, we provide a bounding box of the person who is performing the action indicated by the filename of the image. There are 9,532 images in total with 180-300 images per action class.

TABLE I: Summary of the Datasets

Dataset	#Training	#Test	#Classes	Image Size
Common benchmark datasets				
CIFAR-10	50,000	10,000	10	32×32
CIFAR-20	50,000	10,000	20	32×32
STL-10	105,000	8,000	10	96×96
ImageNet-10	11,700	1,300	10	224×224
Complex scene datasets				
Stanford-40	8,580	952	40	224×224
PPMI-7	700	70	7	258×258

PPMI-7 [62] contains images of humans interacting with seven different musical instruments, including bassoon, erhu, flute, French horn, guitar, saxophone, and violin.

Three widely used clustering metrics including Normalized Mutual Information (NMI), Clustering Accuracy (ACC) and Adjusted Rand Index (ARI) are utilized to evaluate our method. Higher scores indicate better clustering performance.

B. Experiments Setup

Clustering phase: We use BLIP-2 as the caption generator, where each image is input together with an external prompt to generate the corresponding textual caption. Specifically, we adopt the general strategy for common benchmark datasets, using the prompt “a photo of { }”. For complex scene datasets, we employ the customized strategy with the question “What are the people doing in the picture?”. Following [29]–[31], CLIP is used as the encoder for images and text. The clustering head is implemented as a FC layer with a structure of $d_t - K$. In this phase, only the clustering head is trained, using the Adam optimizer with a learning rate of 0.001 for common benchmark datasets and 0.01 for complex scene datasets.

Distillation phase: The image transformation strategy follows SimCLR [22], where two distinct random augmentations are applied to each image before being fed into both the teacher and student networks. The student network adopts ResNet-34 [63] as its backbone. To align the output feature dimensions between the teacher and student models, a FC layer with dimensions $d_s - d_t$ is appended to the backbone. The clustering head is implemented as a FC layer with dimensions $d_t - K$. This phase is optimized using SGD with a learning rate of 0.01, momentum of 0.9, and weight decay set to $1e-4$.

The final experimental results are obtained by averaging multiple runs. The experiments were conducted on a desktop equipped with an Intel Xeon(R) Silver 4310 CPU, an NVIDIA GeForce RTX 4090 GPU, 128GB of RAM, and implemented using the PyTorch 2.0 deep learning framework, Python 3.8 on Ubuntu 22.04.

C. Performance Comparison

Clustering performance: To evaluate the performance of our method in clustering tasks, we compare our experimental results with 11 traditional deep clustering methods, including JULE [64], DEC [20], DAC [59], DCCM [65], PICA [66], SCAN [24], GCC [67], CC [36], IDFD [68], SPICE [26], RPSC [27], as well as 5 externally guided deep clustering methods, including 4 matching methods: SIC [29],

TABLE II: Comparison of clustering performance on four common benchmark datasets

Datasets	CIFAR-10			CIFAR-20			STL-10			ImageNet-10			#Param
Methods	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	NMI	ACC	ARI	
Traditional deep clustering methods													
JULE [64]	0.192	0.272	0.138	0.103	0.137	0.033	0.182	0.277	0.164	0.175	0.300	0.138	0.39M
DEC [20]	0.257	0.301	0.161	0.136	0.185	0.050	0.276	0.359	0.186	0.282	0.381	0.203	1.67M
DAC [59]	0.396	0.522	0.306	0.185	0.238	0.088	0.366	0.470	0.257	0.394	0.527	0.302	0.26M
DCCM [65]	0.496	0.623	0.408	0.285	0.327	0.173	0.376	0.482	0.262	0.608	0.710	0.555	0.43M
PICA [66]	0.591	0.696	0.512	0.310	0.337	0.171	0.611	0.713	0.531	0.802	0.870	0.761	21.32M
SCAN [24]	0.797	0.883	0.772	0.486	0.507	0.333	0.698	0.809	0.646	-	-	-	11.50M
GCC [67]	0.764	0.856	0.728	0.472	0.472	0.305	0.684	0.788	0.631	0.842	0.901	0.822	11.50M
CC [36]	0.705	0.790	0.637	0.431	0.429	0.266	0.764	0.850	0.726	0.859	0.893	0.822	11.77M
IDFD [68]	0.711	0.815	0.663	0.426	0.425	0.264	0.643	0.756	0.575	0.426	0.425	0.264	11.23M
SPICE [26]	0.734	0.838	0.705	0.448	0.468	0.294	0.817	0.908	0.812	0.828	0.921	0.836	43.21M
RPSC [27]	0.754	0.857	0.731	0.476	0.518	0.341	0.838	0.920	0.834	0.830	0.927	0.858	43.21M
UKDC _{distill}	0.835	0.921	0.836	0.587	0.600	0.434	0.897	0.953	0.901	0.892	0.950	0.893	21.55M
Externally guided deep clustering methods													
CLIP [42]	0.736	0.807	0.637	0.487	0.458	0.294	0.899	0.855	0.831	0.965	0.985	0.967	151.28M
SIC [29]	0.847	0.926	0.844	0.593	0.583	0.439	0.953	0.981	0.959	0.970	0.982	0.961	-
TAC [31]	0.841	0.923	0.839	0.611	0.607	0.448	0.955	0.982	0.961	0.985	0.992	0.983	151.81M
MCA [30]	0.850	0.928	0.849	0.606	0.612	0.455	0.955	0.982	0.960	-	-	-	-
IC TC [35]	0.941	0.975	0.947	0.690	0.584	0.429	0.982	0.993	0.985	-	-	-	73.94B
GrandNorm [33]	0.826	0.911	0.815	0.613	0.606	0.436	0.956	0.983	0.962	0.987	0.994	0.987	151.81M
UKDC _{cluster}	0.961	0.985	0.967	0.674	0.645	0.517	0.984	0.994	0.987	0.992	0.997	0.993	4.10B

TABLE III: Comparison of clustering performance on two complex scene datasets

Datasets	Metrics	SCAN	TAC	IC TC	UKDC _g	UKDC _c
Stanford-40	NMI	0.467	0.768	0.848	0.849	0.910
	ACC	0.397	0.632	0.774	0.742	0.848
	ARI	0.272	0.452	0.718	0.686	0.806
PPMI-7	NMI	-	0.738	0.908	0.822	0.961
	ACC	0.632	0.714	0.840	0.800	0.971
	ARI	-	0.583	0.816	0.688	0.934

TABLE IV: Comparison of clustering performance of different distillation methods

Datasets	CIFAR-10			ImageNet-10		
Methods	NMI	ACC	ARI	NMI	ACC	ARI
KD [50]	0.798	0.901	0.797	0.863	0.934	0.860
SEED [54]	0.770	0.884	0.765	0.835	0.917	0.828
CoSS [69]	0.773	0.872	0.738	0.848	0.922	0.834
Ours	0.835	0.921	0.836	0.892	0.950	0.893

TAC [31], MCA [30], GrandNorm [33], and 1 generation method: IC|TC [35]. Additionally, we report the results of performing k -means directly on the features obtained from CLIP's image encoder as a baseline. As shown in Table II, UKDC_{cluster} demonstrates that, compared to traditional deep clustering methods, external information guidance makes it easier to obtain semantic features of image, resulting in higher clustering performance. Compared to externally guided deep clustering methods, whether matching- or generation-based, our method not only constructs a text space more aligned with the clustering task but also achieves superior performance.

In addition, we report the clustering performance of the general text construction strategy (UKDC_g) and the cus-

tomized text construction strategy (UKDC_c) on complex scene datasets, and compare them with three representative methods: the traditional deep clustering method SCAN, the matching-based method TAC, and the generation-based method IC|TC. As shown in Table III, the customized strategy achieves the best performance, benefiting from the explicitly defined clustering objectives guided by task-oriented textual prompts, whereas the general strategy is limited by the absence of such explicit semantic guidance. Traditional methods such as SCAN struggle to capture high-level semantic information in complex images, leading to suboptimal clustering results. Although TAC incorporates external textual information, it lacks precise control over caption generation aligned with clustering targets, thus limiting its semantic expressiveness. While IC|TC leverages a large language model to enhance textual understanding, its heavy reliance on the text modality diminishes overall clustering effectiveness. In contrast, our proposed method jointly leverages cross-modal consistency between image and text representations, effectively improving caption quality and aligning semantic information to achieve superior clustering performance.

Distillation performance: The clustering performance after distillation is compared with various traditional deep clustering methods, as shown in Table II, UKDC_{distill}. Our proposed method outperforms traditional methods across multiple benchmarks. Traditional methods primarily rely on self-supervised learning to mine the intrinsic structure of the data, but their learning performance is limited on small models with constrained representation capabilities. Our method transfers the rich knowledge from the teacher model to the student model through distillation, significantly improving the clustering performance of lightweight models.

In addition, we compared UKDC_{distill} with three knowledge distillation methods using the same experimental baseline. As shown in Table IV, KD [50] uses the true labels as the cross

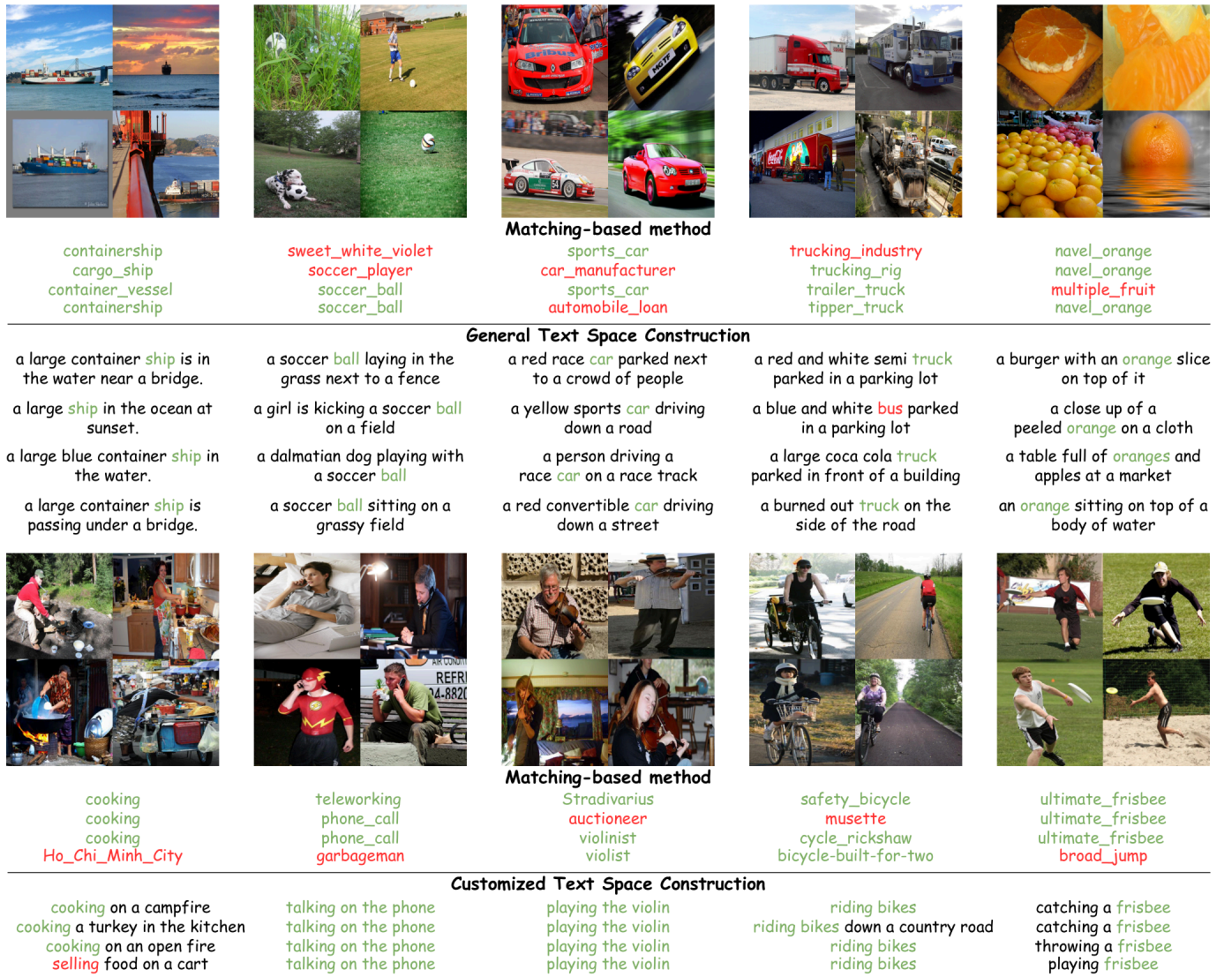


Fig. 3: Examples comparison of different text construction methods. The upper part shows captions for ImageNet-10, and the lower part shows captions for Stanford-40. The text above the horizontal line is constructed using the matching method, while the text below the line is constructed using our method. Green and red indicate whether the text aligns with the image.

entropy loss, and the other methods use pseudo labels instead of true labels. A single distillation strategy is difficult to fully fit the feature distribution and cluster distribution of the teacher model. In addition, the pseudo labels themselves may contain noise, which further affects the clustering quality. In contrast, the feature and cluster dual-level distillation mechanism can more comprehensively align the knowledge structure of the student and teacher models, thereby preserving clustering performance to the greatest extent.

D. Ablation Studies

1) *Effectiveness of text space construction*: Text-image semantic misalignment and intra-text inconsistency are two major challenges faced by matching methods. Semantic deviations are reflected in incorrect matches and the inability of texts to focus on clustering targets, while inconsistencies are reflected in different nouns with the same semantics. As

shown in Fig. 3, in the ImageNet-10 dataset, ‘car’ and ‘loan’, ‘fruit’ and ‘orange’ are incorrectly matched. In the Stanford-40 dataset, clustering is based on human behavior, and the matching method cannot effectively focus on clustering targets, such as ‘phone_call’ and ‘garbageman’, ‘violin’ and ‘auctioneer’. And texts with the same semantics but inconsistent, such as ‘ship’ and ‘vessel’, ‘bicycle’ and ‘musette’, these factors either directly or potentially reduce clustering performance. In contrast, our method is able to not only generate captions that accurately correspond to clustering targets, but also ensure semantic consistency between texts.

2) *Effectiveness of cross-modal consistency learning*: We apply *t*-SNE to reduce the dimensionality of the embedded features and visualize the distributions of text embeddings, image embeddings, and cluster spaces in two-dimensional space. Since multiple images may correspond to the same caption, the number of samples in the text space is less than or equal to the number of samples in the image space, $|\mathbf{T}| \leq |\mathbf{X}|$. As

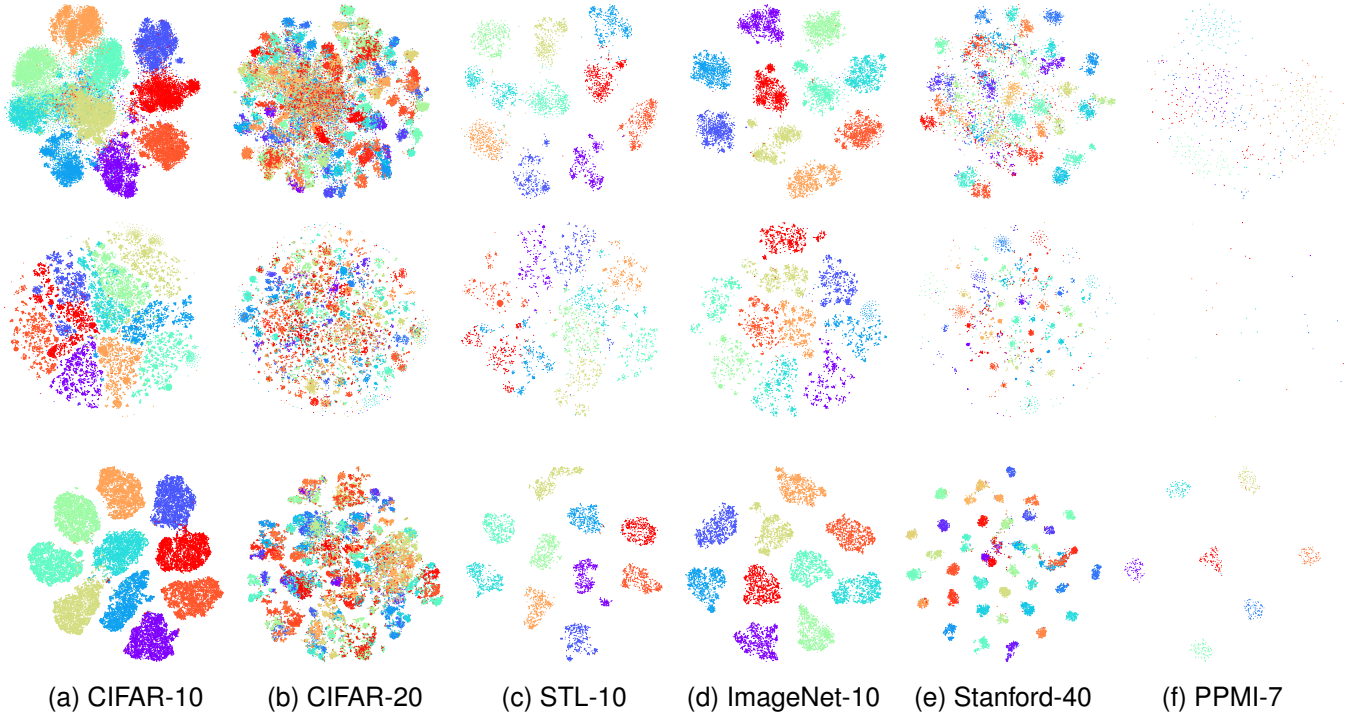


Fig. 4: **Spatial distribution visualization.** The three rows of figures correspond to the image embedding space, the text embedding space, and the final cluster space, respectively.

TABLE V: Ablation analysis during distillation

Datasets	$\mathcal{L}_{feature}$ (w/o) txt $\mathcal{L}_{feature}$ (w/o) img $\mathcal{L}_{feature}$ $\mathcal{L}_{cluster}$	✓	✓	✓	✓	✓
CIFAR-10	NMI	0.568	0.703	0.750	0.805	0.835
	ACC	0.673	0.772	0.833	0.904	0.921
	ARI	0.490	0.649	0.715	0.802	0.836
CIFAR-20	NMI	0.377	0.473	0.506	0.567	0.587
	ACC	0.379	0.477	0.509	0.588	0.600
	ARI	0.220	0.307	0.338	0.415	0.434
STL-10	NMI	0.785	0.858	0.873	0.421	0.897
	ACC	0.812	0.860	0.876	0.585	0.953
	ARI	0.708	0.813	0.830	0.336	0.901
ImageNet-10	NMI	0.720	0.735	0.780	0.858	0.892
	ACC	0.772	0.784	0.802	0.932	0.950
	ARI	0.647	0.659	0.710	0.857	0.893

shown in Fig. 4, on the common benchmark dataset, the image embedding space exhibits compact intra-class distributions, while the text embedding space demonstrates clear inter-class separation, indicating strong discriminative ability. In complex scene datasets, the customized text space construction enables the text space to maintain strong inter-class discrimination and intra-class compactness, even when the image space exhibits limited sample separability. Furthermore, the distribution in the cluster space clearly outperforms both the image and text spaces, indicating that our proposed cross-modal consistency learning method effectively integrates the distribution features of both text and image.

3) Effectiveness of unsupervised dual-level distillation:

Feature distillation and cluster distillation play complementary and crucial roles in the knowledge transfer process. We conducted ablation experiments on Eq.(4) and Eq.(5). When training with feature distillation only, we apply k -means directly on the embedded features instead of using of the clustering head. As shown in the Table V, although cross-modal distillation outperforms intra-modal distillation by reducing redundant information irrelevant to the clustering task, preserving intra-modal distillation can still improve feature properties and achieve better clustering performance. Furthermore, the results demonstrate that both are indispensable. Feature distillation is primarily used to align the representations of the student and teacher models in the embedding space, enabling the student network to capture the discriminative representation of the teacher model. Cluster distillation guides the student model to better learn the distribution structure of the teacher model in the clustering space and understand the relationship between different classes, thereby improving its ability to distinguish between categories. The synergy of the two mechanisms effectively improves the representation and separation capabilities of the student model. It is worth noting that the STL-10 training dataset consists of 5,000 labeled images and 100,000 unlabeled images, where the unlabeled subset contains additional categories beyond the ten target classes¹. In our setting, both labeled and unlabeled subsets are utilized for feature distillation to learn representations. However, for cluster distillation, we only use the clustering assignments generated from the labeled subset to ensure class consistency

¹<https://cs.stanford.edu/~acoates/stl10/>

TABLE VI: Comparison of distillation performance on different student models

Datasets	CIFAR-10			ImageNet-10			#Param
Models	NMI	ACC	ARI	NMI	ACC	ARI	
UKDC _{cluster}	0.961	0.985	0.967	0.992	0.997	0.993	153.39M
Res-18 [63]	0.830	0.920	0.832	0.877	0.934	0.869	11.44M
Res-34 [63]	0.835	0.921	0.836	0.892	0.950	0.893	21.55M
Mob-v3 [70]	0.848	0.925	0.842	0.882	0.949	0.885	3.47M
Eff-b0 [71]	0.850	0.930	0.854	0.898	0.954	0.899	4.67M

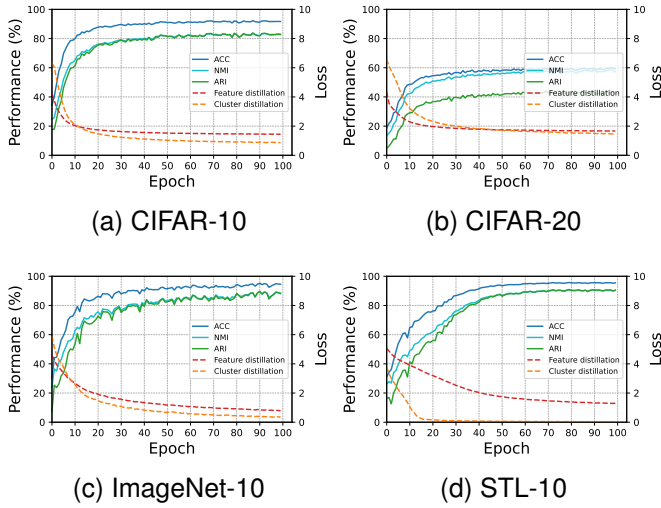


Fig. 5: Convergence analysis

with the test set (Labels are unavailable in all training phases). Since these labeled samples account for only about 1/21 of the total training data, the limited cluster assignments may distort the learned feature structure, which in turn slightly degrades the overall clustering performance.

4) *Distillation results on different student models:* To evaluate the generalization ability of the proposed distillation strategy, we fixed the teacher model and tested it on student models with different architectures. As shown in Table VI, the ResNet [63] model, with only 1/13 the number of parameters of the teacher model, retains at least 86% of its performance. The MobileNet [70] and EfficientNet [71] models, with only 1/44 and 1/32 the number of parameters, respectively, achieve at least 88% of the teacher model's performance. Our dual-level unsupervised distillation method not only effectively reduces the number of model parameters but also preserves excellent clustering performance, closely matching the teacher model's performance across various student architectures.

E. Convergence analysis

We conducted a convergence analysis of the distillation process on a common benchmark dataset. As shown in Fig. 5, all models exhibit a consistent training pattern, with performance metrics improving significantly within the first 20 epochs, while traditional self-supervised clustering methods usually require 500 epochs to achieve this performance. This suggests that the student model can quickly capture the global structure

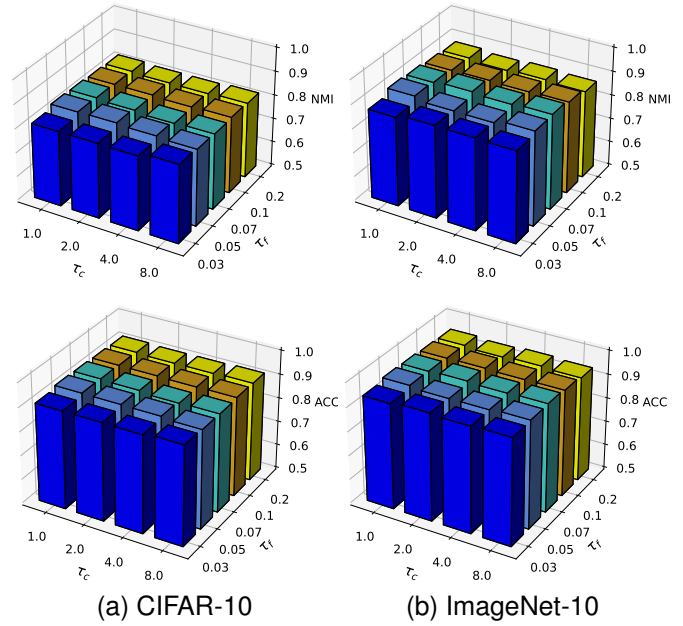


Fig. 6: Effect of hyper-parameters on clustering performance.

of the data under the guidance of the teacher model. As training progresses, the performance curves gradually stabilize, indicating that the models reach a state of convergence. In the early stages of training, the cluster distillation loss is initially higher than the feature distillation loss. However, as training continues, the cluster distillation loss rapidly decreases and the feature distillation loss becomes dominant. On the STL-10 dataset, the cluster assignments of cluster distillation are relatively small. The overfitting of cluster distillation affects the cluster separation, which is consistent with the conclusions in the section IV-D3.

F. Parameter sensitivity analysis

In Eq.(3), the temperature parameter τ_f controls the smoothness of the feature similarity distribution, while in Eq.(5), τ_c regulates the sharpness of the logit distribution during cluster distillation. As shown in Fig. 6, the impact of these two hyper-parameters on model performance remains relatively stable within a reasonable range, indicating that the model exhibits low sensitivity to variations in the temperature parameters. It is evident that the best performance is achieved when $\tau_f = 0.05$ and $\tau_c = 2.0$. The general stability across a broad range of temperature values demonstrates the robustness of the proposed method, which makes it conducive to practical deployment scenarios.

CONCLUSION

In this study, we propose a text-guided unsupervised dual-level knowledge distillation image clustering framework, termed UKDC. This method combines deep clustering with knowledge distillation for the first time and employs VLMs to construct an auxiliary text space for text-guided image clustering. In addition, we introduce a dual-level distillation

strategy that integrates feature-level and cluster-level distillation, significantly reducing the model's parameter size while preserving clustering performance to the greatest extent. Experimental results on six public datasets highlight the advantages of UKDC in clustering performance and model compression, effectively reducing computational and storage costs in real-world deployments. Future research will focus on improving text-guided strategies for clustering in complex scenarios with multiple granularities and perspectives, and addressing the false negative problem of contrastive learning in unsupervised knowledge distillation.

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